

FAKE CHIPS AND CIRCUIT DESIGNING

Recently I read the articles 'Fake Chips: Should Engineers Worry?' and 'Electronic Circuits Have To Be Designed - Not Assembled' that were published in November issue. I didn't know fake chips are there in supply chain. Also, even though I'm interested in designing circuits, most of the time I'm searching the circuits on internet. I agree with the author's bottoms up approach for designing, which is really helpful. Thanks to EFY and the author!

Prakash U.M.

...

DIY FOR ENTREPRENEURS

Instead of the information about successful people and their ups and down, if do-it-yourself (DIY) projects and boards with diagrams are published in detail, it will be more beneficial to the society. These days most of the people are leaving their jobs and switching over to their own business, be it big or small.

I was one of the subscribers earlier but discontinued because of some personal problem, but never went away from you and your publication. Indeed, your magazine is unique and very much needed in the time we are going through. I would like to renew my subscription again, so please let me know the rates for getting the issues through courier.

S.K. Chakravarty

EFY. Thanks for the lovely feedback and encouraging words, which keeps us going. We continue to include DIY projects, big and small, which are a favourite with many. Recently, a design section has been included to help our technopreneurs to tackle on-the-ground challenges and develop innovative products. And in the new section on investment, we would be discussing opportunities for startups

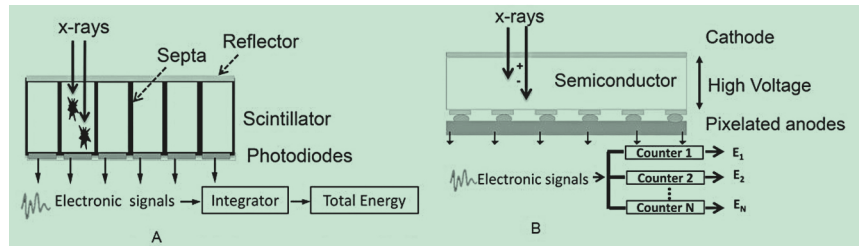


Fig. 1: Illustration of (a) EID and (b) PCD technologies (Credit: pubs.rsna.org)

and technopreneurs, and share challenges and possible solutions. Subscription can be renewed at <https://renew.efyindia.com>

...

MINI PROJECTS

I wish to make a mini project for a contactless switch. Please send me details of the same or let me know where I can find the necessary information.

Eswari Reddy

EFY. The DIY project 'Gesture-Controlled Contactless Switch For Smart Home' is available on our website at <https://www.electronicsforu.com/electronics-projects/gesture-controlled-contactless-switch-smart-home>. Complete description of the project, including source code, can be downloaded from there. Its project kit (components, project description, PCB, etc) can be ordered through our website kitsnspares.com website or by writing to info@kitsnspares.com

...

WHAT IS PHOTON-COUNTING TECHNOLOGY USED IN A CT SCANNER?

Prajil T.

EFY. Photon-counting technology, or photon-counting computed tomography (PCCT), is a form of x-ray computed tomography (CT) in which x-rays are detected using a photon-counting detector (PCD) where it can directly transform x-ray photons into electrical

signals. The PCDs are better than the currently used CT detectors as they provide much higher spatial resolution with improved contrast-to-noise ratio at lower radiation dose, and with intrinsic spectral information.

PCDs do not require a separate layer to convert x-rays into light but consist of a single layer of semiconductor diodes. If an incident x-ray is absorbed in the semiconductor, it generates a cloud of positive and negative charges that get pulled away from each other rapidly. The moving charges generate an electrical pulse in the wires attached to the electrodes, which is registered with an electronic readout circuit.

PCDs thus convert individual x-ray photons directly into an electric signal, unlike energy-integrating detectors (EIDs) used in a conventional CT, which requires the additional step of converting photons to visible light. Fig. 1 illustrates the principles of EID and PCD technologies.

Basically, a conventional CT requires two conversion steps to turn photons into a medical image. Here, x-rays are collected by a scintillator, which uses many photons to generate an optical signal. A photodiode is then used to convert this optical signal to an electrical signal.

This new approach of image acquisition requires an entirely new detector material, a high-purity cadmium telluride (CdTe) crystal for generating a high spatial resolution.

Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: onedit@efy.in)